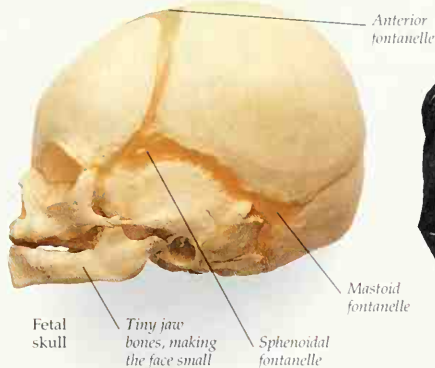
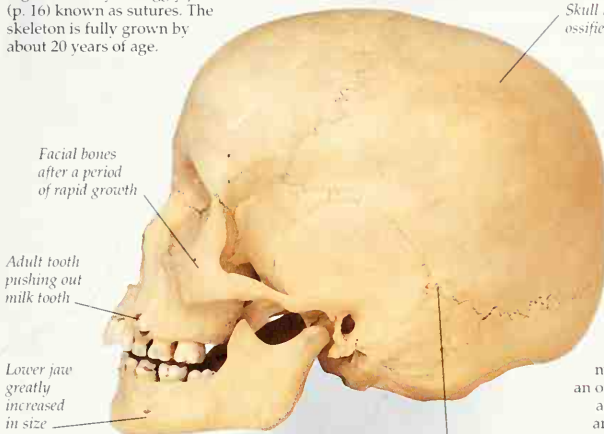


GROWTH AND THE SKELETON

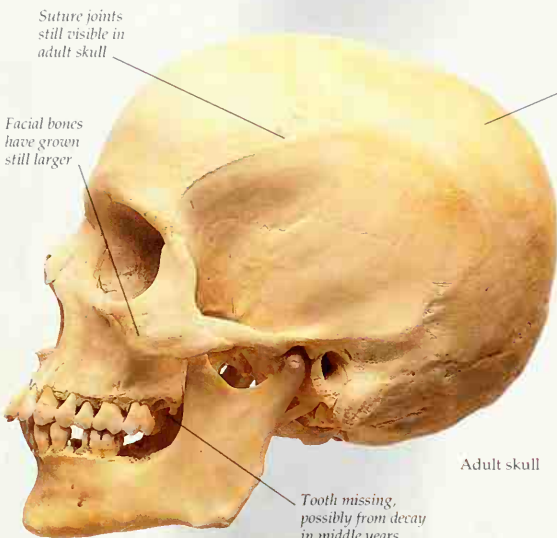
The skeleton first develops as soft, gristly cartilage (p. 19). Bone tissue spreads out from various sites in the cartilage, known as centers of ossification, and turns the cartilage into bone (pp. 16-17). In a newborn baby's skeleton some of the "bones" are still partly cartilage, such as the skull shown here, the wrists, hands, ankles, and feet. The new baby's head is mostly brain. During birth the part-cartilage skull bones (p. 15) move together at soft areas called fontanelles, to mold the skull to the shape of the mother's birth canal (p. 46). The baby's face is very small. In childhood the facial bones grow faster, and the skull bones knit together firmly at wiggly joints (p. 16) known as sutures. The skeleton is fully grown by about 20 years of age.



Skull bones have ossified completely



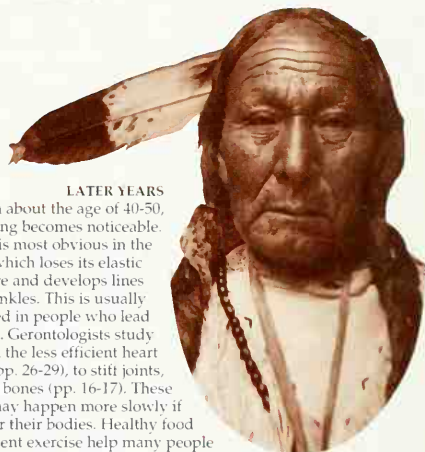
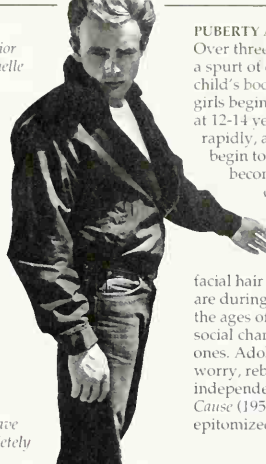
Six-year-old skull



Adult skull

PUBERTY AND ADOLESCENCE

Over three to four years of puberty, a spurt of development changes a child's body into an adult's. Most girls begin at 10-12 years, and boys at 12-14 years. The body grows very rapidly, and sex organs (pp. 42-43) begin to function. A girl's body becomes more rounded, she develops breasts, and she begins to menstruate. A boy's body becomes more muscular, his voice deepens, and facial hair may grow. Puberty years are during adolescence, between the ages of 12 and 20; they involve social changes as well as physical ones. Adolescence can be a time of worry, rebellion and newfound independence. In *Rebel without a Cause* (1955), movie star James Dean epitomized brooding teenage years.



LATER YEARS

From about the age of 40-50, aging becomes noticeable.

It is most obvious in the skin, which loses its elastic nature and develops lines and wrinkles. This is usually most marked in people who lead an outdoor life. Gerontologists study aging, from the less efficient heart and lungs (pp. 26-29), to stiff joints, and fragile bones (pp. 16-17). These changes may happen more slowly if owners care for their bodies. Healthy food and sufficient exercise help many people enjoy good health into their 70s and 80s.



THE LIFE CYCLE

Philosophers through the ages have pointed out that all things are born to die. The human life cycle passes through stages of anatomy and physiology, as well as social and intellectual changes. They include growth in the womb, infancy, childhood, puberty, through adulthood, middle age, and old age. Today, the average life expectancy in many countries is approaching 80 years of age, more than twice as long as in the 1600s. Better public health and improved medical care, food, living conditions, and education can all help people minimize risks and keep their bodies fit and healthy.